

Table 17.3: Critical skills gaps for which no training is available in the country

NDPIV Programme	Field/Skill Area	Critical skills gaps for which no training is available in the country	Estimated 5-year Gaps	Entry-Level Education
Agro-Industrialisation	Agricultural Engineering		538	Bachelor's
	Agricultural Finance Experts		295	Master's
	Irrigation engineers		330	Bachelors
	Quantitative animal breeder		103	Bachelors
	Agricultural Lawyer		970	Bachelor's
	Agricultural Biotechnology Experts		200	Master's
	Agricultural Risk Management Analysts		175	Master's
	Livestock Nutritionists		225	Master's
	Power Grid Maintenance Technicians		125	Diploma
	Process Safety Engineers		290	Master's
	Soil Science		311	Bachelor's
	Veterinary Medicine		400	Bachelor's
Tourism Development	Border Security Analysts		140	Bachelor's
	Tourism Financial Analysts		60	Master's
	Tourism Policy Analysts		150	Master's
	Tourist Service Quality Assessors		60	Master's
	Destination Branding Specialists		75	Bachelor's
	Hospitality Training Specialists		673	Bachelor's
	Tourism Marketing Specialists		300	Master's
Human Capital Development	Chiropractors		652	Master's
	Geriatric Pharmacist		332	Masters
	Geriatric Doctors		240	Masters
	Geriatric Neurologist		150	Masters
	Geriatric Psychiatrist		268	Masters
	Neurosurgeons		109	Master's
	Anesthesiologists		321	Master's
	Advanced Endoscopy		186	Master's
	Optometrists		349	Master's
	Physiotherapists		241	Master's
	Thoracic Surgeons		211	Master's
	Interventional Pulmonologists		145	Master's
	Pulmonologists		354	Master's
	Oncologists (Thoracic/Pulmonary)		121	Master's
	Pulmonary Rehabilitation Specialists		203	Master's
	Allergists/Immunologists		163	Master's
	Critical Care Specialists		361	Master's
	Respiratory Therapists		237	Master's
	Advanced Pediatric Care		350	Master's
	Infectious Disease Specialists		421	Master's
	Virologists		328	Master's
	Cardiologists		350	Master's
	Cardiothoracic Surgeons		124	Master's
	Cardiac Electrophysiologists		105	Master's
	Pediatric Cardiologists		55	Master's
	Cardiac Anaesthesiologists		178	Master's
	Interventional Cardiologists		189	Master's
	Cardiovascular Imaging		400	Bachelor's
	Clinical Cardiology		450	Master's
	Metabolic Medicine		350	Master's
	Nephrologist		300	PhD
	Oncologist		650	Master's
	Radiation Therapists		321	Master's
	Oncologic Pathologists		118	Master's
	Paediatric Oncologists		123	Master's
	Hematologists (Hematologic Oncologists)		250	Master's
	Orthodontists		225	Master's
Sustainable Extractives	Gold Processing Plant Specialists		100	Master's
	Industrial Process specialists		172	Master's
	Machine Learning Engineers		650	Master's

NDPIV Programme	Field/Skill Area	Critical skills gaps for which no training is available in the country	Estimated 5-year Gaps	Entry-Level Education
Industry Development	Mineral Resource Development Specialists		250	Master's
	Pipeline engineers		278	Master's
	Oil Refinery Safety Engineers		225	Bachelor's
	Refinery technicians		570	Bachelor's
	Oil Well Logging Specialists		175	Bachelor's
	Environmental scientists		600	Master's
	Oilfield Quality Control Specialists		225	Master's
	Petroleum Geologists		700	Master's
	Polymer Scientists		140	Master's
	Process Simulation Analysts		47	Master's
	Product Design Engineers		300	Bachelor's
	Underground Mining Engineers		260	Master's
	Environmental Engineering and Policy Analysts		140	Bachelor's
	Geographical Information Systems (GIS)		390	Bachelor's
Integrated Transport Infrastructure and Services	Geotechnical Engineers		155	Master's
	Hydroelectric Plant Engineers		100	Master's
	Infrastructure Investment Analysts		184	Bachelor's
	Materials Scientists		150	Master's
	Traffic Engineering professionals		109	Master's
	Infrastructure Risk Analysis and Management		195	Bachelor's
	Smart Grid Analysts		155	Bachelor's
	Smart Transportation Systems Engineers		255	Master's
	Aviation Experts		350	Master's
Innovation, Technology Development and Transfer	Transport Economists		105	Master's
	Aerospace and Avionics Engineering		50	Bachelor's
	Cyber and Space Technology		50	Bachelor's
	Mechanical/Mechatronics Engineering		50	Bachelor's
	Telecommunications/Antenna Engineering		50	Bachelor's
	Radar Engineering in Space craft systems		50	Bachelor's
	Computer Engineering in Aerospace		50	Bachelor's
	Software Engineering in Aerospace		50	Bachelor's
	Industrial and Manufacturing Engineering		50	Bachelor's
	Remote Sensing and Signal Processing		50	Bachelor's
	Data Science and Analytics in Aerospace		50	Bachelor's
	Embedded Systems in Aerospace technology		50	Bachelor's
	Machine Learning and Aerospace Systems		50	Bachelor's
	Aviation Law and Policy		50	Bachelor's
	Space Law, Policy and Governance		50	Bachelor's
	Space mission project Planning, Monitoring and Evaluation		50	Bachelor's
	Aerospace Material Sciences		50	Bachelor's
	Power Systems/Green energy in Aerospace		50	Bachelor's
	Nuclear Physics and Space mission Biosafety		50	Bachelor's
	Battery Systems Engineer		132	Bachelor's
	Automotive Software Engineer		231	Bachelor's
	Automotive Mechanical Engineer		150	Bachelor's
	Automotive Parts Designer		125	Bachelor's
	Automotive Data Analyst		156	Bachelor's

NDPIV Programme	Field/Skill Area	Critical skills gaps for which no training is available in the country	Estimated 5-year Gaps	Entry-Level Education
Digital Transformation	Charging Infrastructure Technician (Electrical Engineering Technicians)		127	Bachelor's
	Cloud Computing Engineers		900	Master's
	Cybersecurity Specialists		582	Master's
	AI Systems Engineers		202	Master's
	Blockchain Developers		500	Bachelor's
	Cybersecurity Architects		250	Master's
	Digital Forensics Experts		200	Master's
	Human-Computer Interaction Specialists		250	Master's
	Network Architects		600	Master's
	Software Architects		650	Master's
	Virtual Reality Developers		150	Master's
	Industrial Process Consultants		172	Master's
Sustainable Energy Development	Renewable Energy Engineers		200	Master's
	Nuclear Engineering		1,130	Bachelor's
	Nuclear Reactor Operation		240	Bachelor's
	Nuclear Chemistry		630	Master's
	Nuclear Physics		500	Master's
	Nuclear Geology		570	Master's
	Nuclear Safety and Security		1,720	Master's
	Nuclear Law		165	Master's
	Radiation Protection		1,960	Bachelor's
	Nuclear Technicians		1,324	Bachelor's
	Nuclear Craftsmen		3,600	Certificate
	Green Hydrogen Economy		139	Bachelor's

Source: NPA Human Resource Projection Model, 2024

439. The HCD Programme should also incentivise higher education and training institutions to provide the skills needed to fill the human resource gaps in the key growth areas. These are highlighted in Table 17.4.

Table 17.4: Critical skills gaps whose training is available in Uganda but supply is less than the current demand

NDPIV Programme	Field/Skill Area	Critical skills whose training is available in Uganda but supply is less than the current demand	Estimated Medium-Term Gaps	Entry-Level Education
Agro-Industrialisation	Agricultural Machine Operators		1600	Diploma
	Animal Husbandry		4920	Diploma
	Dairy Science specialists		2081	Bachelor's
	Disaster Risk Management		1652	Bachelor's
	Food and Beverage Processing		5280	Diploma
	Food microbiologists		1368	Bachelor's
	Food Science and Technology		3600	Bachelor's
	Food Science Specialists/ Food Scientists		1321	Bachelor's
	Animal Health Specialists		3150	Bachelors
	Veterinary Epidemiologist		1358	Bachelors
	Quality Regulators		475	Diploma
	Bio-Medical Engineering		215	Bachelors
	Laboratory Specialists		1697	Bachelors
	Wildlife medicine Specialists		1212	Bachelors
	Entomologist		727	Bachelors
	Veterinary Doctors		1309	Bachelors
Tourism Development	Event Planning and Management experts		1,360	Diploma
	Hotel Management Specialists		1,930	Diploma
	Adventure Tourism Coordinators		580	Diploma
	Tourism Guides		1,137	Diploma

NDPIV Programme	Field/Skill Area	Critical skills whose training is available in Uganda but supply is less than the current demand	Estimated Medium-Term Gaps	Entry-Level Education
	Adventure Tourism Planners		800	Bachelor's
	Cultural Experience Facilitators		560	Bachelor's
	Cultural Heritage Management Specialists		800	Bachelor's
	Tourism Development Planners		400	Master's
	Tourism Resource Managers		1,137	Bachelor's
	Travel Safety Analysts		580	Master's
	Travel Safety Coordinators		1,881	Diploma
Human Capital Development	Anesthesiologists		521	Master's
	Orthopaedic surgeons		574	Master's
	General surgeons		432	Master's
	Emergency Medicine		561	Master's
	Paediatricians		643	Master's
	Physicians/ Internal Medicine		1590	Master's
	Ear Nose and Throat Surgeons		550	Master's
	Registered Paediatric Nurse		1,091	Diploma
	Registered Public Health Nurse		11,781	Diploma
	Registered Nurse		47,171	Diploma
	Optometrists		749	Master's
	Clinical Microbiologists		213	Master's
	Epidemiologists (Infectious Disease Epidemiologists)		304	Master's
	Pathologists		231	Master's
	Pharmacologists		332	Master's
	Immunologists		227	Masters
	Physiotherapists		541	Master's
	General Practitioners		12,211	Bachelor's
	General Surgery		353	Bachelor's
	Emergency Medicine		1,266	Master's
	Dental surgeons		1,812	Master's
	Registered Midwife		33,754	Diploma
	Nutritionist		2,987	Bachelor's
	Obstetrics and Gynecology		649	Bachelor's
	Clinical Medicine		4,800	Bachelor's
	Early Childhood Education		4,320	Diploma
	Endodontists		3,500	Master's
	Midwifery		6,000	Diploma
	Nutrition and Dietetics		4,200	Diploma
	Occupational Health and Safety		1,137	Diploma
	Occupational Therapy		1,137	Bachelor's
Sustainable Extractives Development	Biochemistry Specialists		1156	PhD
	Bioenergy Specialists		1,050	Bachelor's
	Biotechnology Specialists		1,360	Bachelor's/ Master's
	Environmental Policy professionals		740	Master's
	Petroleum Engineering professionals		543	Bachelor's
	Geophysicists		680	Bachelor's
	Drilling technicians		1,268	Diploma/Technical
	Plumbing Labourers		2,940	Lower Secondary
	Renewable Energy Policy		2,500	Master's
	Renewable Energy Systems		1,137	Bachelor's
Integrated Transport Infrastructure and Services	Renewable Energy Technician		2,920	Diploma
	Pavement Engineers		540	Bachelor's
	Transport Engineers		655	Bachelor's
	Materials Engineers		573	Bachelor's
	Civil Engineers		630	Master's
	Environmental Engineering experts		1145	Master's
	Mechanical Engineers		2135	Bachelor's
	Water and Sanitation Engineering Technicians		755	Bachelor's

NDPIV Programme	Field/Skill Area	Critical skills whose training is available in Uganda but supply is less than the current demand	Estimated Medium-Term Gaps	Entry-Level Education
	Automotive and Power Engineering experts		683	Master's
	Civil Engineering Technicians		850	Master's
	Logistics and Supply Chain Managers		1,184	Bachelor's
	Railway Engineering professionals		1,731	Bachelor's
	Environmental Impact Assessment specialists		919	Bachelor's
	Geotechnical Engineers		865	Master's
	Advanced Roadway Design Engineers		1,105	Bachelor's
	Rail Infrastructure Specialists		1,435	Bachelor's
	Advanced Traffic Management Specialists		774	Master's
	Road Safety Engineers		625	Bachelor's
	Infrastructure Investment Analysts		580	Bachelor's
Innovation, Technology Development and Transfer	Big Data Analysts		450	Master's
	Prototype Engineers		350	Bachelor's
	Data Scientists		700	Master's
	DevOps Engineers		350	Bachelor's
	Digital Data Architects		200	Master's
	IoT (Internet of Things) Experts		450	Master's
	Industry 4.0 Experts		125	Bachelor's
	Robotics Engineers		250	Master's
	Intellectual Property (IP) Managers		155	Bachelor's
	Automotive and Power Engineering experts		183	Bachelor's
	Biomedical Engineering		1,992	Bachelor's
	Automotive Diagnostic Technician		1432	Certificate
Digital Transformation	Biopharmaceuticals		400	Bachelor's
	Digital Marketing		4620	Bachelor's
	ICT and Digital Literacy		4920	Diploma
	Mobile Application Development		4320	Bachelor's
	Network Administration		4920	Bachelor's
	Network Engineering		5400	Bachelor's
Sustainable Energy Development	Software Engineering		1152	Bachelor's
	Electrical Engineering		3554	Bachelor's
	Electrical Technicians		4037	Certificate
	Mechanical Engineering		3240	Bachelor's
	Radiology		900	Bachelor's/ Master's
	Renewable Energy Technician		4920	Diploma
	Renewable Resource Management		1800	Bachelor's

Source: NPA Human Resource Projection Model, 2024

17.6 Implementation Reforms

440. The key implementation reforms required to fully implement this Programme and realise expected goals in the next five years are:

- Address conflict of interest in the provision of education and health services by banning all employees in these sectors from operating private schools and health facilities;
- Children in the day, primary, and secondary schools should attend the nearest schools to their homestead;
- Fees charged in public schools should go through the consolidated fund;
- Introduce a minimum of one year of compulsory TVET training immediately after A' level before enrolling for further education to be delivered through a compulsory National Service programme framework;

- v) In line with the National Human Resource Development Plan the Government issuance of work permits must be based on human resource gaps identified by the plan. In this regard, work permits should be provided in areas where the country is facing staffing shortages;
- vi) Establish and implement a National Central Admission System for higher education and link higher education admissions and financing to the critical skill needs identified in the Plan;
- vii) Link allocation of scholarships and loan financing to critical skill needs identified in the plan; and
- viii) Establish international exchange programmes with multinational corporations.

17.7 Programme Risk and Mitigation

441. **Some of the potential risks that may impede the achievement of the desired results of the programme, along with the corresponding mitigation measures, include:**

- i) **Uganda is faced with a youth bulge of 31.2 million people which creates both an opportunity and a challenge.** The greatest risk factor in this circumstance is that only 15% of the Ugandans that enter the labour market are demanded. The incompatibility between the qualifications possessed by graduates/ those seeking employment and the prevailing demand within the labour market worsens the risk of attaining the desired human capital outcomes. Secondly, the skills and training acquired do not match the market demands in the current job market causing high risk in investments made plus low adoption of technology coupled with emerging artificial intelligence with new technology and innovations; and
- ii) **Increasing vulnerability due to risks and shocks caused by increasing climate changes, pandemics, epidemics, and occupational safety and health risks.** These are a big threat to the attainment of health, education, and productivity outcomes both contributing to mortality and morbidity rates, high cost of health service delivery, destruction of social infrastructure, human resources, and low labour productivity. There is a need to prioritise mitigation, prevention, and management measures for emergencies including refugee influx that impacts on service delivery within the host communities and local governments. In addition, changing climate change contributes to crises e.g., floods, storms, prolonged drought, hunger and poor nutrition outcomes, displacement of people that will require emergency services, disaster risk management, social protection interventions as well as increased communicable and non-communicable diseases leading to high disease burden and high cost of social service delivery.